

Research

A comparative study of deepfake facial manipulation technique using generative adversarial networks

Wasin Al Kishri¹ · Jabar H. Yousif² · Mahmood Al Bahri² · Muhammad Zakarya² · Naveed Khan³ · Sanad Sulaiman Al Maskari² · Ahmet Gurhanli⁴

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Abstract

The manipulation of digital images has become a popular trend. Due to the development of image processing tools and visualization techniques integrating deep learning and artificial intelligence (AI) algorithms (in particular generative adversarial networks–GAN), this can pose serious threats to privacy and security. In recent years, Deepfake algorithms have been designed to exchange faces or modify facial features, potentially leading to more severe problems in this context. In this manuscript, we provide a comprehensive review of the two most important facial image processing technologies: (i) deepfake face manipulation; and (ii) face manipulation detection techniques. Furthermore, we explore the state-of-the-art of popular GAN techniques. In particular, three types of Deepfake face detection technologies are reviewed: (i) hand-crafted features (ii) artifacts; and (iii) learning-based features, while highlighting related improvements and challenges. Furthermore, this article discusses potential challenges and promising research directions for future investigation. We believe that this review has been organized to provide a structured analysis of important research papers and to discuss each study's main findings and conclusions. Our investigation reveals shortcomings in manipulation detection benchmarks due to real-world scenario variations and biased dataset comparisons. Current research priorities revolve around enhancing GAN training stability, resolution, and manipulable facial features. Moreover, GANs have shown superior results in identifying fake images; however, their reliance prompts a systematic approach to detecting fakes. This dependency raises questions about detecting fake images with or without manipulated GAN architecture, urging the need for novel computational techniques to identify manipulations without GAN assistance.

Keywords Deepfake · Artifacts · Deepfake generation · Deepfake detection · GAN applications · GANs

1 Introduction

Nowadays, the manipulation of digital images has become a popular trend. With the new development of image processing tools and visualization techniques, many people post flawless facial images on social media and use filters to appear different. On the other hand, DeepFakes illustrates computer graphics and visualization techniques to manipulate

Jabar H. Yousif and Mahmood Al Bahri have contributed equally to this work.

✉ Wasin Al Kishri, wasan.k@aou.edu.om; ✉ Muhammad Zakarya, mzakarya@su.edu.om; ✉ Naveed Khan, n.khan@ulster.ac.uk; ✉ Sanad Sulaiman Al Maskari, smaskari@su.edu.om; ✉ Ahmet Gurhanli, ahmetgurhanli@topkapi.edu.tr; Jabar H. Yousif, jyousif@su.edu.om; Mahmood Al Bahri, mbahri@su.edu.om | ¹Faculty of Computer Studies, Arab Open University, Seeb, Oman. ²Faculty of Computing and IT, Sohar University, Sohar, Oman. ³Department of Computing, Ulster University, Belfast, UK. ⁴Department of Computer Engineering, Istanbul Topkapi University, Istanbul, Turkey.



and replace the face of a person with another [1]. The term 'DeepFakes' refers to a method based on deep learning and originated in 2017 by Reddit after he developed a machine learning algorithm that helped him discredit celebrities by turning their faces into pornographic videos fake news, hoaxes, and financial theft are some of the most malicious applications of such fake content [2]. As a result, a new research field is devoted to forensics in the general media [3]. The researchers are dedicating growing efforts to detecting fake face images in two directions, which are previous studies in biometric anti-spoofing [4, 5], and current data-driven deep learning [6, 7]. There are various reasons for focusing on the facial information parameter. Firstly, faces are a vital part of human communication, as a person's face can send or even express a message in its own right [8]. Secondly, human face reconstruction and tackling are well-examined fields in computer vision [9]. The availability of the public data set and the advanced development of deep learning techniques make it easy to change and manipulate a person's real face in an image or even a video.

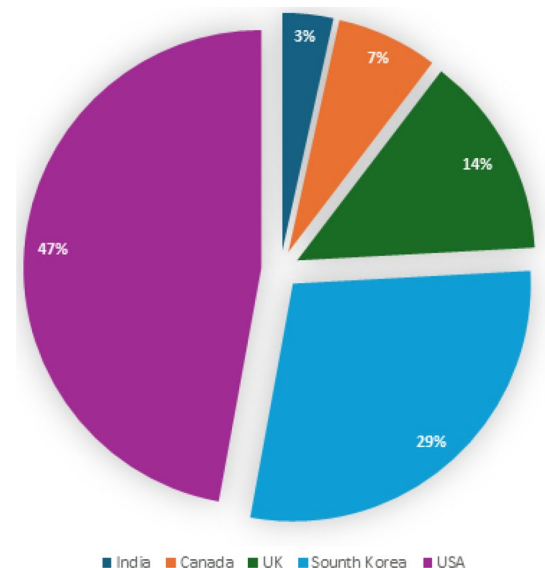
Face manipulation consists of the face changing in the image's particular features of the face, such as the hair color or the eyes, the gender, the size, the appearance of glasses, etc. Deepfakes have started to appear everywhere. DeepFake technology is a double-edged sword that can be used positively and negatively. We will begin with some positive uses of DeepFake technology. In the face of the COVID-19 pandemic, when shooting corporate training videos with actual actors is becoming more difficult and expensive, [10] uses DeepFake technology to provide corporations with cost-effective synthetic training videos. DeepFake is also used to help alter the visual effects of the face for film and television productions, recreate the appearance of certain celebrities who may have died, or modify their appearance by removing facial features such as wrinkles or adding some sparkle to the face. DeepFake also has the ability to bring together celebrities across geographic and generational boundaries in a creative landscape. For example, [11] used DeepFake technology to broadcast Dutch celebrities on live TV, including Prime Minister Mark Rutte and Queen Maxima of the Netherlands. Nowadays, most social media applications (Instagram, Snapchat, etc.) change the user's appearance or entertainment. From viral celebrity face swaps to impersonations of political leaders, it can be hard to spot the difference between real and fake. The world is entering an era of evolution, technology, and artificial intelligence, which makes Deepfake so easy for anyone to say anything at any time. Fake news, hoaxes, and financial theft are some of the most malicious applications of such Deepfakes content. The use of artificial intelligence in Deepfakes makes it difficult to discover the authenticity of photos or videos and thus affects the public's trust in information sources and the belief in what they see [12–17].

In this paper, we offer a comprehensive review of various approaches that have been suggested and implemented in state-of-the-art research related to deepfake face manipulation and face manipulation detection techniques. Furthermore, we deeply investigate the current research related to GANs and their role in DeepFakes detection. Although GANs have demonstrated better performance in recognizing phony images, their dependence necessitates a methodical approach to false detection. This dependence highlights the need for new computational methods to detect manipulations without the use of GANs, raising concerns regarding the detection of fraudulent images both with and without altered GAN architecture. The major contributions of this paper are:

- offers a thorough analysis of deepfake facial modification methods with a particular emphasis on generative adversarial networks (GANs). Complete face synthesis, identity switching, attribute manipulation, and expression swapping are the four main face modification techniques that are categorized;
- investigates a number of deepfake detection strategies, including deep learning-based approaches, artifact-based detection, and handcrafted features. It addresses the efficacy of GAN-based detection systems and draws attention to the difficulties in identifying altered information;
- identifies key issues in deepfake research, including the need for better training stability, dataset biases, and GAN fingerprinting; and
- makes recommendations for future research paths, such as more reliable deepfake detection models and methods that do not only use fingerprints and other approaches based on GANs.

The rest of the paper is organized as follows. In Sect. 2, we discuss various issues related to DeepFakes and elaborate on why it is essential to find practical solutions. The organization of this study, papers collections form published sources, and selection criteria is deliberated in Sect. 3. In Sect. 4, we discuss various Deepfake face manipulation methods using GAN. In Sect. 5, we deliberate on face manipulation detection approaches. Different applications of GAN within the context of DeepFake are described in Sect. 6. A discussion on the obtained outcomes of this study is reported in Sect. 7. Finally, we conclude this paper in Sect. 8.

Fig. 1 DeepFake pornography websites



2 Deepfake issues

In this section, we present some of the threats (particularly in the finance sector) of Deepfake implementation that make it necessary to find practical solutions for them.

2.1 Targeting deepfake in financial sector (theft)

The digital impressions are starting to have real financial repercussions. According to a new report in The Wall Street Journal,¹ in the U.S., an audio deepfake of a CEO reportedly scammed one company out of 10 to transfer €220,000 (approx. \$243,000) to the bank account of a Hungarian supplier.

2.2 Targeting deepfakes in pornographic films

In 2017, Reddit developed a machine learning algorithm that helped in discrediting celebrities by turning their faces into pornographic videos [2]. In 2019, Amsterdam-based Deeptech conducted research to discover the prevalence of fake videos on the Internet.² The report noted that 96% of fake online videos were of a pornographic nature, with only 4% depicting topics other than porn, such as politics or humour. Comparing between 2018 and 2019, the report published a 100% increase in the prevalence of fake videos online, as 25% of porn movies are from Korean singers, while 41% of videos are from American celebrities [18]. While websites target images of women 100% for making porn movies, YouTube channels target images of women at 39%. So, for that, we need a detector to differentiate between real and fake faces, as shown in Fig. 1.

2.3 Deepfake source code on github

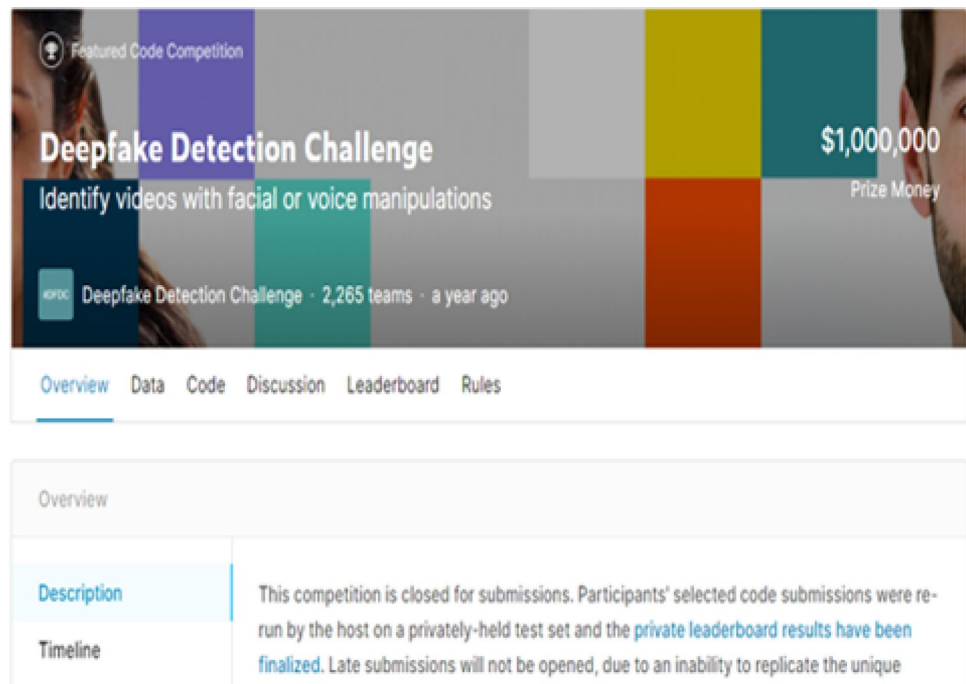
It is an open-source community³ where code and machine learning projects are uploaded to create Deepfakes for researchers and programmers. Furthermore, the repository has been extensively used where programmers find loopholes, improve the use of proposed programs, and increase their efficiency and quality. Most researchers [7, 19, 20] used unsupervised GAN-based frameworks (generative adversarial networks) for translations and manipulations of digital images. In addition, on the practical side of facial attribute manipulations [20–22]. The figure below shows the number of research papers in which the GAN approaches have been published from 2014 to 2019, and they have already achieved

¹ <https://threatpost.com/deep-fake-of-ceos-voice-swindles-company-out-of-243k/147982/>.

² https://regmedia.co.uk/2019/10/08/deepfake_report.pdf.

³ <https://github.com/topics/deepfake>.

Fig. 2 A view of the Website for Google's Deepfake detection challenge



excellent results in various scenarios. This manuscript focused on the new direction, which is a GAN-based framework removal, to find the manipulation.

2.4 Competition for deepfake detection using prize

In 2019, Google launched a program to advance Deepfake detection, inviting researchers to submit countermeasures against deepfakes. Now, tech giants like Google, Twitter, and Microsoft are stepping up. Facebook was spending more than 10 million dollars to fight Deepfake, as shown in Fig. 2.

In the computer vision community, the study of DeepFake and, in particular, the use of GAN techniques to create fake faces for whatever reason has certainly gained traction among researchers in recent years. Figure 3 shows the number of published scientific papers on DeepFakes and GANs technologies that have increased from 2016 to 2023. It also shows a significant increase in the number of papers and the increasing interest of researchers in the topic of DeepFakes. Also, it reviews the latest literature related to Gan's approaches to raising awareness of the emerging threat of DeepFake technology.

This paper reviews recent articles on facial manipulation techniques using generative adversarial networks (GANs). This review discusses different GANs' face architecture, evolution, and various advantages and disadvantages of using GANs for DeepFakes. This paper's main objective is to closely examine how the most popular GAN algorithms are applied to facial content for digital manipulation techniques. The most serious common security threats against DeepFake and the most important reasons for addressing and combating this phenomenon were reviewed. Specifically, four main types of face manipulation algorithms using GAN techniques are covered:

1. Entire face synthesis,
2. Identity swapping,
3. Attribute manipulation, and
4. Expression swapping.

From the oldest to the newest, all the negatives and positives of the four classifications of generating fake faces using GANs techniques, methods, datasets, code links, and accuracy are discussed in detail. This, in turn, will help researchers to take an overview of the evolution of the algorithms and the improvements that have been added. Besides these face manipulation techniques, this review covers several challenging and trending applications used to generate DeepFake face manipulation. The recent and public databases for research are introduced in ascending

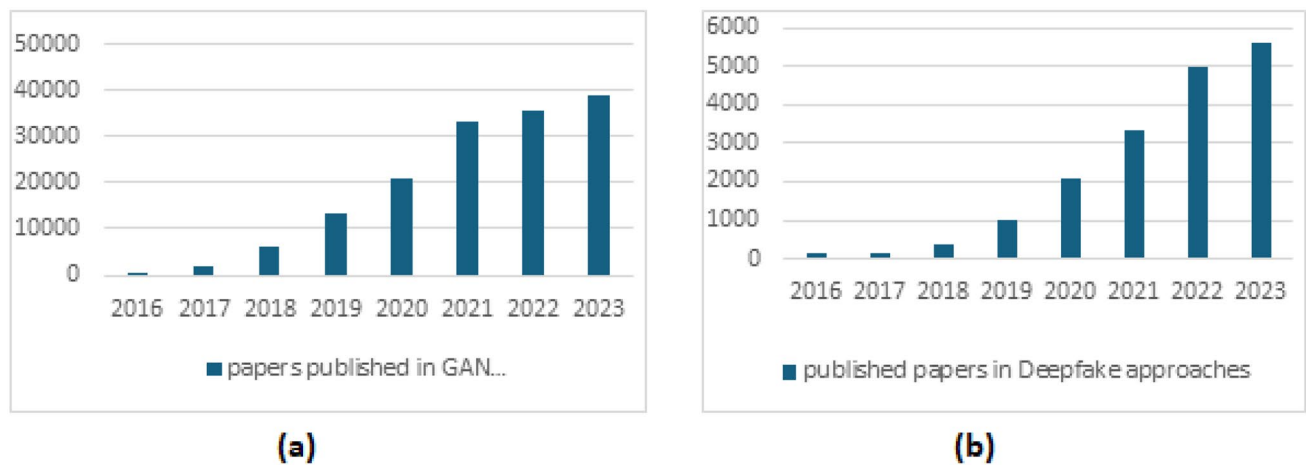


Fig. 3 The number of research papers in which the GAN (a) and DeepFake approaches (b) were published in Google Scholar from 2016 to 2023. According to the figure, the increase is expected to continue in the coming years

order by their release dates and the purpose for which they have been implemented. The categorization of various DeepFake detection approaches and their challenges is described in this paper as shown in Figure 4. In our final remarks, we highlight future trends and open questions for further research.

3 Paper collection and scope

This manuscript reviews two of the most important technologies for facial image processing: (i) deepfake face manipulation; and (ii) face manipulation detection techniques. In order to collect various published articles, we used keywords such as “deepfake face manipulation”, “Face Manipulation Detection Techniques”, and “GAN”(generative adversarial networks) to search for relevant studies. The inclusion criteria for selecting the papers in this review were the year of publication (2017–2023), subject, study relevance, and impact of the study in the field of deepfake face manipulation and face manipulation detection techniques. These papers were collected from different sources, including Google Scholar, arXiv, ScienceDirect, Springer, IEEE, and GitHub (for source code). Our research focused on identifying recent research papers published in scientific journals and conferences, aiming to uncover the latest advancements in the field of Deepfake Face Manipulation and Face Manipulation Detection Techniques. We collected a total of 132 papers using various keywords and then based on the abstract study, we chose 100 references for our paper, including 42 papers from indexed international conferences (IEEE/CVF, PAMI, ECCV, and AAAI) and 58 from indexed journals. The papers from indexed international conferences covered a wide range of topics related to deepfakes, including face manipulation,

Fig. 4 A generative adversarial network architecture for DeepFakes

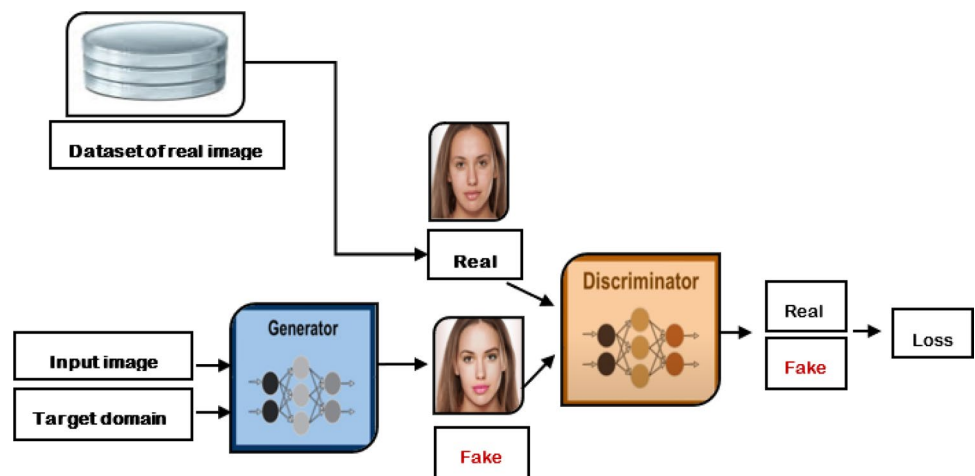
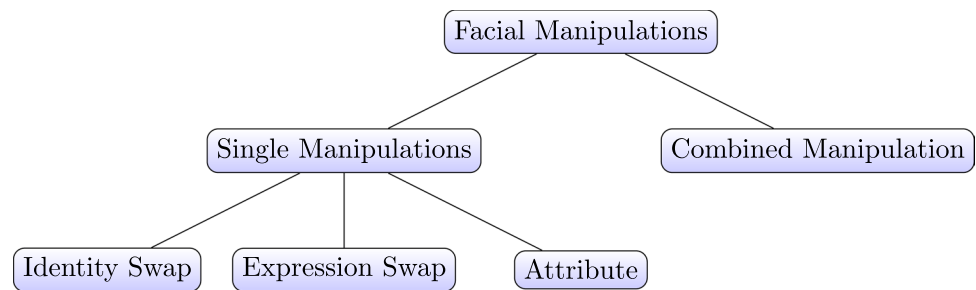


Fig. 5 A taxonomy of various face manipulation approaches



deepfake detection, and the potential impact of deepfakes. The papers from indexed journals provided a more in-depth look at the technical aspects of deepfakes, including the use of deep learning and generative adversarial networks.

The references that we collected provide a comprehensive overview of the state of the art in deepfake research. They will be used to support the arguments that we make in our paper and to provide evidence for our claims. We specifically sought papers from indexed high-impact factor journals and international conferences, which undergo rigorous review processes by expert reviewers. By analyzing these papers, we aimed to gain insights into the current state of the art, emerging trends, and innovative approaches, architectures, and algorithms in deepfake face manipulation and face manipulation detection methods.

4 Deepfake face manipulation methods using GANs

4.1 Types of facial manipulations

Facial manipulations can be classified with regard to the degree of manipulation in four different key categories, as shown in the taxonomy provided in Fig. 5. Three of them belong to a single approach, while the fourth one belongs to a combined approach. These types are: (i) identity swap—replacing the identity of the person; (ii) expression swap—altering facial expressions while keeping the identity intact; (iii) attribute manipulation—changing attributes like age, gender, hair color, or other facial features; and (iv) combined manipulation—applying multiple manipulations together, such as (i), (ii), and (iii). A breakdown of each of them is provided below, ordered from the highest to the lowest manipulation level.

4.1.1 Identity swap

This type of manipulation involves changing one person's face in a video with another person's face. There are two different methodologies: (conventional computer graphics-based techniques, such as FaceSwap and DeepFakes, which are novel deep learning techniques, for example, the latest smartphone application of ZAO. You can see authentic videos of this type of manipulation on YouTube and commercial websites. For example, many industries may benefit from this form of exploitation, including the film industry. On the other hand, it can be for passive exploitation and wrong reasons, such as creating popular pornography films, hoaxes, and financial fraud in different ways.

4.1.2 Expression swap

This modification, also known as face re-enactment, involves manipulating the individual's facial expression. Various manipulation techniques are suggested in the literature, for example, through popular GAN architectures at the image level [20]. These methods usually concentrate on the most popular Face2 Face techniques and neural textures [23]. Furthermore, these approaches help to replace the outward facial expression of an individual in a video with the external facial appearance of someone else. This kind of control could be utilized with real outcomes, but at the same time, it can be used to exploit severely bad results.

4.1.3 Attribute manipulation

This type, also otherwise called face altering or face correcting, comprises changing some ascribe and features of the face, for example, the shade or colour of the hair, eyes, and skin, the sex, the age, including glasses, facial scars [24]. This method is typically brought out through GAN, for example, the StarGAN approach proposed in [25]. The famous FaceApp mobile application is one example of this form of manipulation. This technology could be used by customers to try on a wide variety of items in a simulated world, such as cosmetics and lipstick, glasses, or hairstyles.

4.1.4 Entire face synthesis (combined)

This type of manipulation creates whole nonexistent face pictures, typically through incredible GAN approaches, for example, through the StyleGAN method suggested in [26]. These approaches produce expected outcomes, delivering high-quality facial images with a high degree of realism. The exploitation here could help many fields, such as the video game and 3D modeling industries. However, it could also be used to create disinformation for applications, such as creating realistic fake social networking profiles.

Deepfakes have gained popularity because of the high quality of the manipulated videos. There are widespread advanced free tools to manipulate the features of face images without the need for programming skills. Most of these apps are developed based on deep learning algorithms as presented in Table 1. Machine learning and artificial intelligence techniques are the main components of Deepfake technology, allowing it to be created faster and at a lower development cost. The developer trains a neural network on many hours of genuine video to give the neural network a realistic “knowledge” of what the real person looks like from different angles and under varying lighting. The trained network is then used with computer graphics methods to replace the person’s copy with another target person [27]. Generative adversarial networks (GANs) are the most popular deep learning methods that can detect and automatically learn patterns in input data to generate new examples that can be derived from the original dataset.

4.2 Generative adversarial networks (GANs)

Generative adversarial network, or, in short, GAN. GAN was first invented by Ian Goodfellow in 2014, and since then, these models have seen an incredible burst of research improvements and applications [28]. Many researchers used a fully unsupervised GAN architecture to translate and manipulate digital images. This is because the resource code of most GAN approaches is open-sourced by their creators. GAN modules discover the underlying structure in that dataset, and once the model has discovered the structure, it allows for exploring the variety of features. GAN includes two main neural networks, which are the generator and the discriminator. The generator oversees the following:

1. Randomly sampled noise vector as an input in most cases.
2. Feed the input through a bunch of convolution layers.
3. Produce the output image, so the generator tries to create an image that fools the discriminator.

This output image is fed to a second neural network called a discriminator. The discriminator has one job, which is to look at an image and decide whether that image comes from an actual dataset of the real image that was trained on or whether that image came from the generator and is thus a fake. Using these label classification (real, fake) images allows backpropagating of a training loss through this discriminator network to make it better. Then, the generator network is a fully differentiable neural network that allows learning signals through the pipeline model to update a single loss function in both the discriminator and generator networks and give better results, as depicted in Fig. 6. Despite the scientific development and the production of fake images using the GAN algorithm, many defects appeared. The researchers have, therefore, come up with many improvements to this algorithm that produce a realistic image that matches the real image better in terms of quality, diversity, and stability [29].

4.2.1 Deepfake generation using GANs techniques

The following section presents a review of Deepfake face manipulations using GANs architecture. With the appearance of the GAN and its ability to create realistic images, it has gained a lot of attention from scientists [30–33]. There are many common failure modes associated with the GAN generator and discriminatory architecture in terms of training and

Table 1 A review of applications that have been developed based on Deep Learning algorithms

Applications	Description	Advantages	Disadvantages
DeepFaceLab: https://github-for-windows.en.softonic.com/	Make high-fidelity false videos that seem real	A powerful deepfake software for PC. Rich features to replace the face, de-age the face, replace the head, and manipulate lips	Not easy to use for beginners
DeepVideo Portraits	It uses GAN to turn input video into photo-realistic video	More lifelike effects Swap full 3D head position, rotate head, transfer eye gaze, and blinking. Mimic the facial expressions, eye gaze, and blinking	The code is not available
FakeApp https://www.malavida.com/en/soft/fakeapp/	Using artificial intelligence to copy the face attribute target to someone else	Create deepfake videos on Instagram	Not easily used by non-tech people
Toongineer Cartoonizer https://vanceai.com/toongineer-cartoonizer/	It uses AI and deep learning to convert video into deepfake images and video	Make deepfake images online with AI. Intuitive interface Offers a free trial	Not able to alter face with others. The deepfake feature is limited to images
ZAO Deepfake App https://zaodownload.com/	It works with bout OS Android and iOS, it is used to transfer one face features to another person	A straightforward deepfake video app. Compatible with Android and iOS. Video editing features. Free deepfake app	Involve privacy risks
Reface https://hey.reface.ai/	It creates and produces realistic deepfake videos in one tap	This AI deepfake application is quite simple to use. In a single swipe, transform anyone into a movie star	Not free
Disney's Deepfakes	It can generate deepfake videos with a high-resolution reach (a 1024 x 1024)	Produce high-quality deepfake images	There is no demo product
DeepFake Bot https://github.com/dome272/Instagram-DeepFake-Bot	This enables the creation of deepfake videos on Instagram and the conversion of source footage into animated motion sequences	Use Instagram to post deepfake videos	Inaccessible to non-techies
MachineTube	It creates deepfake images or videos	Free deepfake software	Need a computer has a system with high standard. Produce movies of low quality

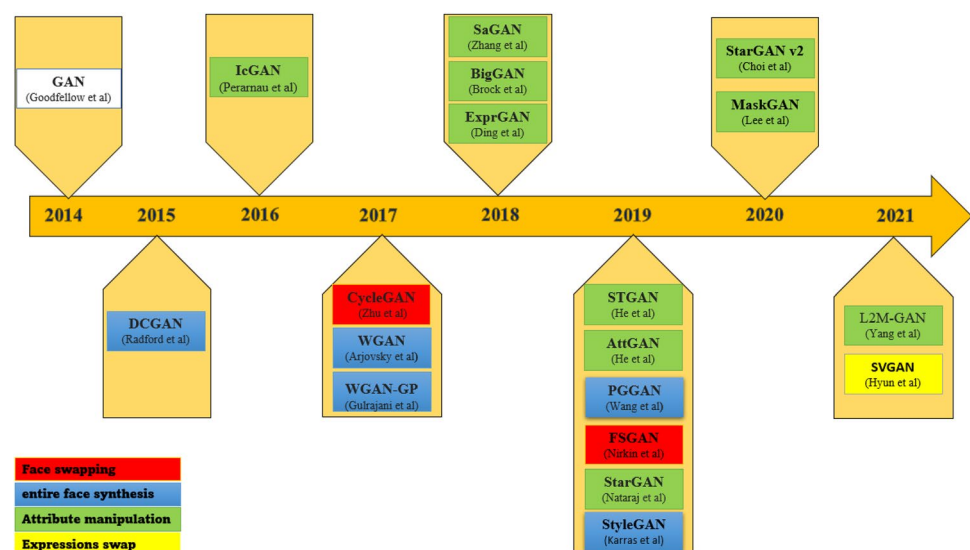
evaluation. The problems are the subjects of current research. That's because researchers continue to find improved GAN technologies. For example, different studies focus on enhancing training stability [34, 35], and producing high-resolution images by adding starburst patterns on their own in necessary areas instead of fetching them from the original image [36]. Regardless, Conditional GANs train on specified labeled data, or unconditional GANs train on unspecified labeled data [19]. To overcome the lack of paired training data, CycleGAN [37] proposes using cycle consistency loss. StarGAN [38] and StarGAN v2 [39] handled multidomain image translation instead of a single domain. Inpainting GANs were used in the semantic picture mapping task. The system attempts to fill in the parts of the image that are blurred during the inpainting process [40]. MaskGAN enables users to edit a file by using the semantic mask interface to change the shape and add accessories such as glasses. Also, many other studies have been deployed to improve the quality of GANs [41].

The study conducted in [42] shows that the first work combining CNN and GAN is the deep convolutional generative adversarial network (DCGAN). In image classification tasks, it is comparable to the pre-trained discriminator in terms of its unsupervised learning ability. The generator is capable of manipulating many semantic properties (for example, the attributes of a human face). Invertible conditional GAN (IcGAN) [43] is one of the first algorithms used to modify and reconstruct some facial features of real human face images. Then other algorithms appear to edit photo-realistic facial expressions and selective attributes with controllable expression intensity, such as selective transfer GAN (STGAN) [21] expression GAN (ExprGAN) [44]. Arjovsky et al. [45] proposed WassersteinGAN (WGAN), which only adds a few optimization designs to the original GAN so as to achieve a reasonable and efficient approximation of the expectation-maximization (EM) distance. In [34], the authors showed that WGAN could produce poor samples or lack convergence due to its use of weight clipping to enforce a Lipschitz constraint. In order to improve this issue, they proposed implementing WGAN-GP (gradient penalty WGAN) to penalize the gradient of the discriminator in relation to its input fake image.

In [46], the authors presented BigGAN as a method for generating different images at high resolution using datasets such as ImageNet [47]. The authors used orthogonal regulation to force the generator to accept a simple solution. By adjusting the contrast of the generator input, the user can regulate the trade-off between image resolution and diversity. In [48], the authors proposed a spatial attention GAN (SaGAN) approach to address irrelevant changes of an attribute in the desired face. Changes will only affect the attribute-specific areas of SaGAN, while all other areas will remain unchanged. However, while the face traits are important, requiring that the latent representation be invariant to the attributes is excessive. Therefore, simply constraining the latent representation is harmful to attribute change, since it limits its ability to represent and risks information loss.

As an improvement to the SaGAN module [49], the authors proposed the facial attribute editing (AttGAN) module, in which the strict attribute-independent constraint is removed from the latent representation. The attribute classification constraint is applied to the resulting image to ensure that the attribute manipulation was done correctly. In order to achieve high-quality facial attribute manipulation, the system integrates attribute classification, reconstruction, and adversarial learning techniques. The model allows for direct attribute intensity control on various face attribute edits within a single model.

Fig. 6 The timeline evolution of Deepfake generation using GANs techniques



In [50], authors suggested a new approach called FakeSpotter, which specifically monitors the behavior as features of neurons layer by layer in-depth, offering new insight into the discovery of complex artificial intelligence using binary classifiers to recognize real versus fake faces. The researchers applied their experience to detecting GANs (PGGA, NSTGAN & StyleGAN) techniques for synthesizing fake faces. They achieved strong results in detecting the entire face treatment using PGGAN, which obtained 94.8% accuracy. They found that PGGAN is a stable approach to training GAN models to generate large, high-quality images. However, detecting fake videos is challenging, especially since some high-quality fake videos employ several unknown techniques in datasets.

In [51], the authors proposed a new training methodology for generative adversarial networks, which is the progressive growth of GANs for improved quality, stability, and variation. The main idea is to gradually develop both the generator and the differentiator by starting from a lower resolution, adding new layers that design increasingly finer details, such as Training progress. This significantly speeds up and stabilizes training, resulting in high-resolution images. They used the CELEBA dataset and achieved an accuracy rate of 80.8%. Face swapping GAN (FSGAN) for face swapping and re-enactment by [52]. Using RNN-based face re-enactment, the authors develop an approach that adapts for variations in position and expression. They introduce re-enactment, Delaunay Triangulation, and barycentric coordinates for face interpolation. Their final step is to combine the two faces seamlessly while maintaining the target skin colour and lighting conditions. Perceptual loss is combined with Poisson optimization in this network. However, there were several limitations in the facial re-enactment experiment due to significant face angle differences, further degrading identity and texture quality. Additionally, many iterations of the facial re-enactment generator result in blurred textures. Also, since the background was not removed from the facial images, another limitation was the inability to capture complex facial expressions.

Reference [41] introduces MaskGAN, a novel framework for diverse and interactive face manipulation. Their main finding is that semantic masks are an excellent intermediate representation for flexible face alteration while maintaining fidelity. The dense mapping network (DMN) and editing behavior simulated training are the two primary components of MaskGAN (EBST). DMN learns style mapping between a free-form user-modified mask and a target image, allowing for a wide range of generation results. EBST models user editing behavior on the source mask, making the whole framework more resistant to varied manipulated inputs. It introduces dual-editing consistency as an auxiliary supervision signal, in particular. A unique feature of MaskGAN is that users can edit a file by using the semantic mask interface to change the shape, location, and class of facial components within the geometric plane. For example, adding accessories such as glasses, earrings, and hats, or the ability to add hair or modify facial features such as the shape of the face, eyes, and nose. To make large-scale investigations easier, they created CelebAMask-HQ, a large-scale high-resolution face dataset with fine-grained mask annotations. MaskGAN has been thoroughly tested on two complex tasks: attribute transfer and style copy, and it outperforms other state-of-the-art approaches [53].

In [39], the authors proposed StarGAN v2, which overcomes two fundamental issues in image-to-image translation: (a) translating an image from one domain to a variety of pictures from a target domain; and (b) supporting multiple target domains. According to the results, their model outperformed the competition in terms of generating images with rich styles across many domains. The authors examine their StarGAN v2 framework under two scenarios: (i) latent-directed synthesis; and (ii) reference-directed synthesis. Using latent vector synthesis, they succeeded in changing the hairstyle of the source images in a manner that takes a lot of time and effort, such as creating high-resolution ear shapes in different shapes. Additionally, the reference-directed synthesis method also renders distinctive styles such as beards, and hairstyles, while other studies mostly match the colour distribution of the reference images.

Reference [54] suggested the manipulation of facial attributes based on the L2M-GAN model. An end-to-end GAN model for editing face attributes based on latent space factors is effective for local and global attribute editing. Because of the new style converter that transforms the latent code into attribute-related and related parts by forcing orthogonality before and after transformation. In a recent study [55], the authors proposed a new algorithm for editing facial expressions called self-supervised video GAN (SVGAN). First, it focuses on the GAN discriminator. The authors then propose two significant constraints for realistic videos: (i) appearance consistency; and (ii) motion coherence. These constraints increase the likelihood that GANs will produce realistic videos. They have clearly established the constraints that synthesized videos must initially adhere to.

Table 2 summarizes various applications developed using deep learning algorithms, highlighting their advantages and disadvantages. Each application, including StyleGAN, StarGAN v2, and others, showcases unique capabilities in image synthesis, yet they all face challenges such as training complexities and issues with image realism, underscoring the ongoing need for improvements in this field.

4.3 Deepfake datasets

DeepFake generation methods' performance tests are often conducted on fake image/video datasets. By developing these methods, the quality of generated fakes has been improved, as has the fidelity of fake datasets. The fake datasets are generated by using real-life video datasets. In order to train DeepFake detectors, real datasets are required. In the first part, we will discuss the popular real-world datasets and then fake datasets.

In [56], the authors proposed a CASIA-Web real-face dataset for collecting a lot of images of faces from the Internet in a semi-automatic way. The dataset contains 10,575 identities and 494,414 images. In [57], the authors proposed a real face data set named CelebA that was generated by classifying images taken from the well-known data set CelebFaces [58]. CelebA produces 200,000 attribute manipulation images from 10,000 identities, each with 20 photos. By a professional labeling company, forty-five face traits and five crucial elements have been tagged on each image of CelebA. Similarly, [59] proposed a VGGFace real dataset with 2622 identities and 2.6 million images that were trained using the VGG Net dataset. In [60], the MegaFace dataset, which consists of more than 690 K unique faces, was released. MegaFace offers a challenge for studying how face recognition algorithms perform under the influence of a large number of distractors.

In [61], the authors introduced the real VGGFace2 dataset, which consists of 3.31 million images with 9131 unique identities. The images were collected from the Internet and vary widely in terms of age, lighting, ethnicity, and occupation. In [62], the authors released the actual data set of human faces known as Flickr-Faces-HQ (FFHQ) with very high accuracy as it is distinguished from others by the ability to add more accessories such as glasses and sunglasses, hats, etc. It contains 70,000 high-quality images with a resolution of 1024×1024 of different age groups, ethnicities, and backgrounds.

In [63], the authors introduced the UADFV dataset that contains 98 videos, 49 actual YouTube videos, and 49 synthesized videos generated using the FakeAPP [64]. As introduced by [65], Google DFD has approximately 3000 modified movies with 28 performers in diverse scenes. Hundreds of real videos are used to create the videos, utilizing publicly available DeepFake generating methods. One of the most famous fake video datasets is called FaceForensics++ [1]. It has 1000 original videos that have been manipulated with the most famous face manipulation tools such as DeepFakes, Face2Face, FaceSwap, and NeuralTextures. They are mostly front faces and, therefore, easy to use in the detector.

In [66], researchers suggested the Diverse Fake Face Dataset (DFFD), a large-scale dataset containing various forms of face forgery. There are 47.7% males and 52.3% females in all images and video frames, and the ages of the samples are between 21 and 50. They employ real face samples from FFHQ, CelebA, and FaceForensics++. FaceForensics++ was used to create the facial identity and expression swaps. For synthesis, they used PGGAN and StyleGAN [51]. FaceAPP [64] and StarGAN [67] were used to modify the face attributes.

In [6], researchers proposed iFakeFaceDB, a huge fake face video dataset that allows for model training and analysis of synthetic face alteration detection. It includes approximately 87,000 synthetic face images created with the StyleGAN model [26] and modified with the GANprintR [6] method. All of the images are 224×224 pixels in size. In [63], the authors introduced Celeb-DF, a large-scale fake dataset that includes the challenging DeepFake video dataset [68]. It contains 5639 high-resolution DeepFake videos of celebrities, created with a synthesis process.

The DeepFake Detection Challenge (DFDC) dataset was proposed in [69]. It is a freely accessible face swap video dataset with 128,514 videos and over 100,000 total frames taken from 3426 celebrities. They employ various face-swapping techniques in a video, such as colour transforms, frame rate changes, etc.

In [70], the authors attempted to create a huge fake dataset called ForgeryNet. It creates a Deepfake face (image/video) classification, spatial forgery localization, and temporal forgery localization. This fake dataset is the largest of the fake datasets, having 2.9 million images and 221,247 videos. It also has 15 manipulation ways with over 36 mix-perturbation options. The OpenForensics dataset was introduced by [71]. This large dataset can be used for detecting and segmenting multi-face forgeries. It is built with face-wise rich annotations specifically for face forgery detection, as presented in Fig. 7.

Datasets that contain abundant taxonomic information, especially facial traits, are superior to attribute manipulation. Among the most widely used datasets for real face attribute manipulation are CelebAHQ and CelebA, as illustrated in Fig. 8. Table 5 shows that CelebAHQ and FaceForensics++ are the two datasets that got the highest citations by researchers and are the most commonly used for full-face synthesis detection and attribute manipulation detection, as presented in Fig. 9.

Table 2 A summary of various applications developed based on Deep Learning algorithms

GAN	Advantages	Disadvantages	Used
StyleGAN [38]	Offers high-quality and realistic pictures even of 1024×1024 resolution	However, existing techniques still have training challenges (for example, mode collapse problems with GAN training procedures). In addition, a lack of basic understanding of facial distribution (such as facial symmetry) makes the generated images insufficiently realistic	Entire face synthesis techniques
StarGAN v2 [39]	Scalability across many domains the diversity of produced images		Attribute manipulation and expression synthesis
PGGAN [50]	Stable approach to training GAN models to generate high-quality images and resolution	Some of the generated images are not real due to insufficient training	Entire face synthesis techniques
DCGAN [42]	Its generator has numerous semantic features that it can control. (example, a human face's attributes)		Entire face synthesis techniques
WGAN [45]	Compared to DCGAN, it produces more realistic and higher-quality synthetic faces	Solves the primary training issues of GANs with permissive restrictions on the discriminator and generator balance, but it requires more training time	Entire face synthesis techniques
SaGAN [48]	Address irrelevant changes of an attribute in the desired face. SaGAN is only focusing on changing the attribute-specific region and keeping the other area unchanged	However, while the face traits are important, requiring that the latent representation be invariant to the attributes is excessive. As a result, simply imposing the attribute-independent constraint on the latent representation limits not just its ability to represent but also risks information loss, which is detrimental to attribute editing	Face attribute
AttGAN [49]	Only the altered attributes are relevant to the editing operation, rather than all of the target properties	Produces poor results and makes unwanted changes, such as blurriness	Face attribute
BigGAN [46]	Creating several high-resolution images	Image synthesis based on class leakage and class-conditional	Face attribute
MaskGAN [41]	Users can edit a file by using the semantic mask interface by changing the shape, location, and class of facial components within the geometric plane. For example, adding accessories to the original images	Images generated by the generator could still be fuzzy and lose some details	Face attribute

Fig. 7 The number of datasets cited by the search on 4 April 2022 from Google Scholar dataset score according to the Elo rating

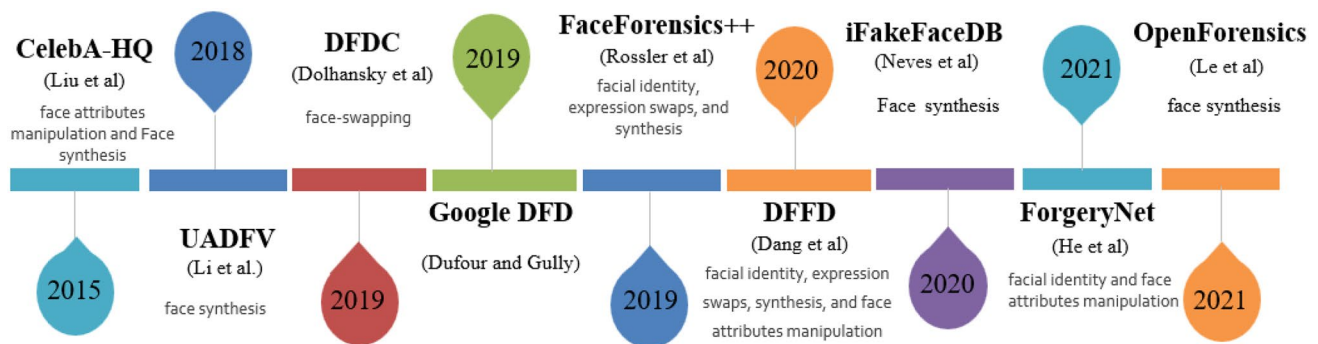
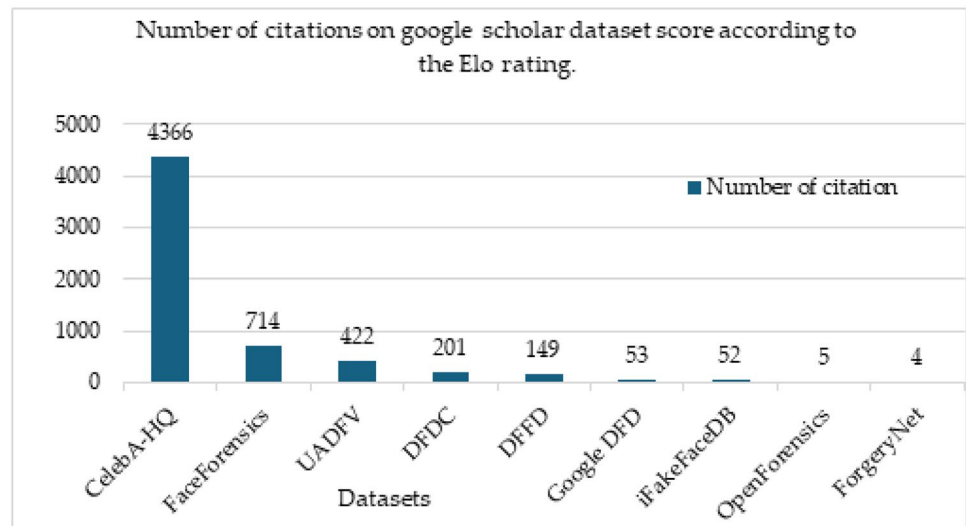


Fig. 8 Timeline evolution of Deepfake generation GANs techniques on different types of face manipulation

5 Face manipulation detection

This section discusses the three Deepfake detection techniques shown in the taxonomy presented in Fig. 9. Also, it explores the currents and their challenges as illustrated in Table 3.

5.1 Hand-crafted features

It uses various algorithms and methods designed by expert designers to extract specific information and characteristics from the face image. For example, detecting the eyes' status (open/closed) can be done by extracting Hand-crafted features (edges and corners) from face images. These are the famous hand-crafted methods for face recognition systems: Gabor filter, local binary pattern (LBP), local ternary pattern (L.T.P), and histogram of oriented gradients (H.O.G).

Robinson et al. [72] conducted the handcrafted (LBP) method and CNN based on VGG-16 for deep feature face extraction. The results of this study show that hand-crafted feature extraction achieves better results in detecting presentation attacks than VGG-16, with 3.1% APCER and 25.2% BPCER.

Fig. 9 A taxonomy of face manipulation detection techniques

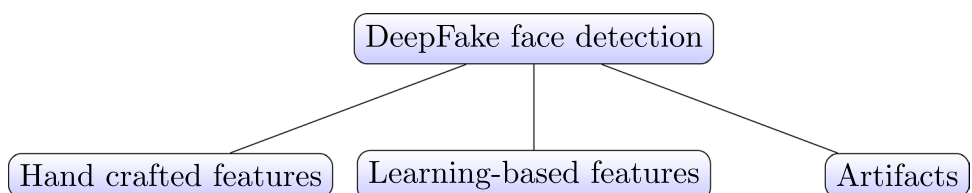


Table 3 A summary of different Deepfake face detection techniques

Deepfake face detection techniques	Description	Advantage	Disadvantage
Hand-crafted features	They are designed with the expert knowledge of designers to extract specific information (edges and corners) and characteristics from the face image. Famous methods are local binary pattern (LBP), local ternary pattern (L.T.P), and histogram of oriented gradients (H.O.G)	It achieves good results in presentation attacks	The extracted features do not correctly represent the characteristics of the problem, resulting in low accuracy and differing accuracy based on the factors of the attack face picture
Learning-based features	Deep learning of the input data, distributed representation of the image. It has a strong focus on learning the inherent characteristics of data sets. (Famous methods (CNNs) to learn the required features	Improve learning efficiency and easily find invariant facial features at any angle or position	It has a slow training speed and is sensitive to noisy data, leading to low final detection accuracy
Artifacts	It is looking for the evidence that can be analyzed to deduce whether it is manipulated or not	Usability and robustness in the detection of Deepfake	Suppose Deepfake detection techniques specialize in features, for example, the irregular blinking rate in a video. In that case, individuals may try to create improved Deepfake methods of production that do not have those characteristics

Agarwal et al. [73] hypothesized that “everyone has facial expressions and special movements while speaking”. Hence, the authors propose a novelty detection model to distinguish a person of interest (P.O.I) from others based on a class SVM method. The authors studied one video clip for 10 s. It is used as an input to track head movements, extract expressions, and distinguish features from the face. They used the OpenFace2 toolkit to extract head movement, brow raiser and lowered lip part, jaw drop, nose, cheek, and eye blink.

Jung et al. [74] proposed the DeepVision method to analyze the eye blinking patterns of the person and detect Deepfakes in the video. The researchers extracted the facial region of the target person and tracked their eye blinking based on calculating their E.A.R. (eye-aspect-ratio) for each frame. They use the E.A.R value to detect the eye’s blinking events (closed/opened) and analyze a person’s blinking threshold for each framework formulation in the video. They achieve an 87.5% accuracy rate.

As a result, the extracted features do not correctly represent the characteristics of the problem, resulting in a low accuracy and differing accuracy based on the factors of the attack face picture.

5.2 Learning-based features

The analysis of learning-based methods for detecting Deepfakes is addressed in this section. Many Deepfake detection strategies, for example, use neural network models (CNNs) to learn the required features.

In the FaceForensics++ article, Rossler et al. [1] test multiple learning and one hand-crafted attribute extraction methods on their ability to detect forgery at various video quality levels. They preprocess the input image (crop and resize on those processes) and then use different methods to spot fakes. All of the methods achieved a high degree of precision on raw input data. However, when watching compressed images, output fell marginally. XceptionNet obtained the best score across all video performance levels among the checked processes. Some methods for making Deepfakes do not produce clear videos temporarily. Artifacts due to this difference may provide good clues to detecting Deepfakes based on the video. One method for incorporating temporal and spatial data into the model’s training is to use 3D CNNs.

In particular, I3D, 3D ResNet, and 3D ResNeXt were investigated by Wang et al. [75] for their capacity to detect altered video.

Sun et al. [76] recommend using optical flow in order to detect deep counterfeiting. The general concept is that there might be variations in movement through frames in deepfake videos (such as irregular movements of parts of the face) that can be picked up by the visual flux. By using the PWC-Net model, the forward optical flow is determined for $(f(t), f(t+1))$, which shows how different objects in a scene appear to move at a given time $f(t)$. The luminous flux values extracted are converted to the RGB image format and used by Flow-CNN to detect deep falsification.

According to Atoum et al. [77], two-stream CNN could provide a new anti-spoofing solution to the problem. According to the proposed method, local characteristics and comprehensive depth maps can be gleaned from pictures of people’s faces. Using the Patch-based approach, random features are selected from the face. On the other hand, the depth features help to make the most of the whole face. The researchers then combined the two methods to discover face spoof patterns. Based on the expected depth map, the facial picture was given an ‘aliveness’ value. The combined values of the two CNNs give the final estimated spoofing class. There was an accuracy rating of 89%.

5.3 Artifacts

Lukáš et al. [78] proposed two approaches to detect fake images using sensor pattern noise. In the first approach, the user chooses a part of an image that needs to be tested for image integrity. The second approach automatically evaluates the forged region without presuming any information. The fake area is described as lacking the noise pattern, while the presence of the noise is calculated using correlation. Photo response non-uniformity (PRNU) is a flaw in the production process of every camera that leaves a distinct mark on the acquired image. The absence of PRNU, as noted by the authors, may suggest tampering. One of the many downsides of these methods is that they require a large number of images to produce accurate predictions and cannot be applied to all synthetic methods.

Tariq et al. [79] present a neural long short-term memory (LSTM) based residual network (CLRNet) for capturing temporal information in consecutive frames. It aims to improve standardisation capabilities and combat fraudulent videos generated using various synthetic techniques.

The study conducted in [80] demonstrates that, like real-world cameras, each GAN leaves its fake fingerprints in the photos it generates. The authors offer an experiment that demonstrates the existence of GAN fingerprints by considering three GAN architectures, Cycle-GAN, ProGAN, and Star-GAN, with larger-scale source identification tests.

Firstly, assume X_i is an image produced by a given GAN. Secondly, measure the residual noise R_i as $R_i = X_i - f(X_i)$, in which f is a denoising filter value.

Thirdly, calculate the random noise component (W_i), and the GAN fingerprint F , $R_i = F + W_i$.

The authors investigated two distinct GAN architectures: (a) a cycle-GAN that has been trained to convert images like an orange into an apple; and (ii) a progressive-GAN trained to produce photographs of kitchens. According to the meaningful image fingerprint correlation, one should be able to determine which GAN produced the given image. After conducting the experiments, they found that Cross-GAN correlations, which measure the relationship between the fingerprint of a GAN and the image produced by another GAN, are uniformly spread about zero, suggesting no correlation between them. Same-GAN correlations, which measure the relationship between the fingerprint of one GAN and the image produced by the same GAN, are slightly greater than zero, showing a strong correlation with the correct fingerprint.

Table 3 summarises Deepfake face detection techniques with advantages and disadvantages.

6 GANs applications

Generative adversarial networks, often known as GANs, are a form of deep learning model that, by learning from a training set, are able to generate data that has the appearance of being realistic, such as images, music, and text. GANs are made up of two different neural networks, a generator and a discriminator, which are trained together in a process that is similar to playing a game of competition. Although the discriminator works to tell the difference between real and fake data, the generator attempts to create fake data that is designed to seem exactly like real data. During the course of time, the discriminator gets better at recognising phony data, while the generator learns to provide data that is increasingly accurate to the real world. The applications of generative adversarial networks (GANs) have grown dramatically over the past few years. This technique has been successful in applications, including high-fidelity natural image synthesis, data augmentation, and improving image compression, among others. GANs can emote super-realistic expressions, explore deep space, and bridge the gap between humans and machines. This section will introduce some of GAN's applications. See Table 4.

Many applications of GANs have been found in fields such as image processing and computer vision, natural language processing, and automatic gesture identification. The following are some examples:

- *Image enhancement*: GANs have the ability to generate lifelike pictures of people, animals, and other objects. The generation of realistic faces is a common use of GANs, and it has applications in a variety of fields, including gaming, entertainment, and even security. For instance, NVIDIA's StyleGAN2 is able to produce synthetic images of human faces that are of high quality, diverse in appearance, and can be controlled.
- According to Khan et al. [81], the authors developed adversarial Gaussian denoiser framework that inspired a new form of GAN algorithms. The main objective of this method is to process blurry images and reduce image noise problems without affecting the realism and quality of the images. By using the proposed model, the denoiser network finds a distribution of sharp images rather than blurry images. Their experiments were conducted on the Partial-CelebA and DIV2 K datasets. As a way of evaluating their method's effectiveness, they compared it to Dn-CNN, FFDNet, and ID-MSE-FFDNet. It achieved high average scores for PSNR, SSIM, UQI, and VIDEO for all noise levels compared to other conventional methods.
- *Features filter*: Nie et al. [84] presented a new model for translating images called GiGAN. Their model considers image-to-image translation to be a sequential job, in which source domain features are gradually transferred to the target-domain equivalents; hence, they employ a gate mechanism to establish sequence correlation. To filter and combine features, the proposed residual blocks with gate cells are used. The GiGAN was able to enhance image quality, as demonstrated by experiments and a performance evaluation.
- *Image manipulation*: According to Ko et al. [82], the authors developed a geometric approach for determining the latent representation and camera pose of 3D GANs based on a single image. Their method can be widely applied to 3D-aware generators by combining hybrid optimization with additional encoders that are trained by manual pseudo-datasets. Through this pre-training session, the encoder will acquire representation and geometry knowledge of 3D GANs, eventually leading to a stable optimisation path. In addition to this, they make use of several different loss

Table 4 A summary of GAN applications with brief description

Study	GAN application	Application description
[81]	Image enhancement	Process blurry images and reduce image noise problems without affecting the realism and quality of the images
[82]	Image manipulation	Geometric approach for determining the latent representation and camera pose of 3D GANs based on a single image
[83]	Video manipulation	Reconstruct and edit 3D faces
[84]	Features filtering	Image-to-image translation by transferring the features to the target image
[85]	Object recognition	Ear recognition based on unsupervised deep active learning generative adversarial network (GAN) to colour the images of the USTB2 dataset, which in turn added more accurate detail and higher quality
[86]	Automatic gesture identification	They proposed a new idea for automatic gesture recognition based on Deep Generative Adversarial Networks (DCGAN), which improves the reliability of trained models. With the help of Convolutional Neural Networks, they train a gesture recognition model
[87]	Natural language processing	GANCoder is made to turn natural language text descriptions into programming code
[88]	Voice translation and audio style transfer	In voice translation, the discriminator distinguishes between the target language speech and the synthesized speech generated by the generator. In voice style transfer, the generator uses the learned style to create a new audio recording that is then compared to the original and tries to be differentiated by the discriminator

functions and provide much improved results in both the quantitative and qualitative aspects of reconstruction and picture integrity. Their research raises the intriguing prospect of utilizing 3D GANs as a form of editing software. It is hoped that this will spur additional study on 3D GAN inversion, which will be utilised further with the single-view 3D reconstruction and semantic attribute modification.

- *Video manipulation:* Xu et al. [83] developed a method for 3D-aware inversion and editing of face video using GAN. In this method, the out-of-distribution objects are isolated from the in-distribution objects in order to reconstruct 3D faces. Through the rendering process, they get an accurate reconstruction by building two radiance fields and composing them together. By modifying the latent code in the distribution part, they were able to obtain plausible results for editing.
- *Object recognition:* Object recognition is a computer vision problem involving the identification and classification of items in an image or video. Using a generative adversarial network (GAN), a form of neural network design consisting of two parts, a generator, and a discriminator, is one technique to approach this challenge. One way to utilize a GAN to recognize objects is to teach the generator to make pictures of different objects from different angles and points of view. Then, the discriminator can be taught to put these photos into separate groups of objects. Once the GAN has been trained, the generator can be used to make fresh pictures of things that the discriminator can use to classify.
- A Different efficient use of the GANs algorithm was described by Khalidi et al. [83], who proposed ear recognition based on unsupervised deep active learning. They used a generative adversarial network (GAN) to colour the images of the USTB2 dataset, which in turn added more accurate detail and higher quality. Therefore, this dataset got a high accuracy rate of 100% compared to the rest of the datasets used, such as mathematical image analysis (AMI), which got 98.33%, and ears annotated web (AWE), which got the lowest accuracy level of 51.25%.
- *Automatic gesture identification:* Deep generative adversarial networks (GANs) are used in automatic gesture identification to recognize and categorize hand movements in real-time using a method known as machine learning. As a result, communication becomes more natural and intuitive. A generative adversarial network (GAN) learns the underlying patterns in a dataset and then uses that knowledge to produce new data. A generator and a discriminator are the two primary components that make up this device. In gesture recognition, the generator is taught to make new pictures of hand gestures, and the discriminator is taught to tell the difference between real and fake pictures of hand gestures. By repeatedly training these two parts, GANs can learn to make realistic images of hand gestures.
- Fang and Ding et al. [86] have demonstrated another application of using GANs. They proposed a new idea for automatic gesture recognition based on deep generative adversarial networks (DCGAN) and CNN. The researchers collected ten real images from their webcam. These images include expressions and gestures. They processed the combined images due to the scattering of light. They generated 44,400 images by applying the DCGAN method. For training, 37,000 images are used, while for testing, 7400 are used. DCGAN generates diverse gestures, which improves the reliability of trained models. With the help of convolutional neural networks, they train a gesture recognition model. With an average recognition accuracy of 90%, they achieved good recognition results.
- *Natural language processing:* An AI chatbot is one is a computer program that simulates a dialogue with users, particularly through the Internet. Various programming languages and technologies can be used to create chatbots, which often use natural language processing (NLP) algorithms to understand user input and respond properly. GANs can be used to produce more human-like responses for chatbots by analyzing enormous datasets of human conversations. This can enhance the chatbot's capacity to comprehend natural language and produce relevant responses [89]. Zhu et al. [87] developed a novel method in 2019 called GANCoder. This method uses generative adversarial networks to automatically translate from natural language to a programming language (GAN). GANCoder is made to turn natural language text descriptions into programming code. This makes it easier for those who don't know much about programming languages to make programmes. The paper talks about how GANCoder is built and how it works. It talks about the training process and the evaluation criteria that are used to measure how well it performs. The results of the tests showed that the GANCoder did better than the baseline models in terms of code quality, accurate syntax, and correct meaning. On the Python function generation challenge, for example, GANCoder got an accuracy score of 85.4%, which was a lot higher than the scores of the baseline models. On the Java method creation task, GANCoder also did better than the baseline models, getting an accuracy score of 82.1%.
- *Voice translation and audio style transfer:* Pasini et al. [88] demonstrates how to construct and train a system that is capable of carrying out audio transmutation as well as any other type of audio pattern transfer (for example, converting one music genre into another). The approach is highly influenced by recent research in image-to-image translation utilizing generative adversarial networks. The primary distinction is that all of these techniques are applied to audio data rather than visual data. They were able to translate samples of arbitrary length, which is an ability that

GAN systems do not typically possess. They utilised the MelGAN-VC model for training the data and separating spectrogram samples, sending them to generator G, stringing them back together, and feeding the generated samples to discriminator D. This allowed for the translation of samples' arbitrary length without any discrepancies occurring. To set the identity in jobs that also require preserving the high-level information of language information in the case of transliteration, an additional constraint that can be used instead has been provided as an optional option.

7 Discussion

This section discusses the obstacles and prospects of the current Deepfake generation and Deepfake detection methods, as well as the potential research directions that could be pursued.

7.1 Deepfake generation challenges and future research directions

In the last few years, Deepfake generation methods have changed a lot. There are four main categories: whole face synthesis, attribute manipulation, identity swap, and expression swap. The high quality of the generated images has made it very difficult for humans to distinguish between real and fake. Using GANs to synthesise an entire face is a form of distribution mapping. The GANs learn the mapping between random distributions and the distribution of human faces. Existing state-of-the-art methods can reliably generate high-resolution images, facilitated by the GAN network's and training procedure's continuous improvement. Despite this, current practices remain challenged by a lack of training (e.g., mode collapse in GAN training). Furthermore, the generated images lack realism due to general information about the face distribution, for instance, facial symmetry. Furthermore, the generated images lack authenticity due to a lack of available information about face distribution, for example, facial symmetry. In entire face synthesis, the fake images are extremely realistic, making it very difficult to tell and identify the difference between real and fake images. Most current research focuses on improving training stability, resolution, and controllable face attributes. Deep convolutional GAN (DCGAN), Wasserstein GAN (WGAN), progressive growing GAN (PGGAN), and StyleGAN. Manipulation of attributes using GANs is a form of latent space editing. One generator will preserve some areas while omitting others. We refer to seminal works in attribute manipulation, such as IcGAN, StarGAN, AttGAN, and STGAN, where the fundamentals of multidomain transformation, accurate attribute manipulation, and new representations for modified attributes were developed over time. Identity swap, the fake videos have uneven qualities. Current work mainly focuses on improving images' realism and resolution, separating the source and target images' attributes and identities, and on how to fuse them. Furthermore, identity swap focuses on improvements in generated images' realism, resolution, and heterogeneous aspects.

Existing works concentrate primarily on increasing the variety of input sources and the consistency of the video. SVGAN is a well-known example of the term expression swap. However, as noted from previous studies, most current face manipulation algorithms seem to be easy to detect under controlled scenarios and using a GAN architecture, as they achieve high detection accuracy and low error rate. It is still an open question of whether GANs leave a stable fingerprint on the generated image datasets. [90] investigates which image components contain fingerprints using different GAN architectures (ProGAN), such as in [51] and CramerGAN [91]. The results demonstrate that GANs have unique model fingerprints and leave stable fingerprints in the produced GAN images. This study obtained a final false detection accuracy of 99.5% for best performance. However, this approach is not very robust against unseen conditions like noise, blur, cropping, or compression. In [92], the authors developed a novel method for detecting deepfake images created by GAN algorithm techniques (StarGAN, GDWCT, AttGAN, StyleGAN, and StyleGAN2). By using discrete cosine transform (D.C.T) to detect anomalous frequencies and not requiring a lot of computing resources for training, the researchers proved that these GAN methods leave unique fingerprint frequencies in generated images. Robustness experiments were also conducted to demonstrate the technique's efficacy in utilizing various assaults on pictures, including JPEG compression, mirroring, rotation, and scaling. The results obtained with FaceForensics++ have a high accuracy of 99.9%. We also want to highlight that most researchers create their own datasets because they lack high-quality publicly available datasets.

Here, we can raise a question: *is it possible to dispense with GAN fingerprints or add some noise patterns while maintaining the realistic aspect of the manipulated image?* These issues motivate the researchers to conduct additional research into the generalization capability of the fake detectors. The researchers presented various preliminary scientific studies in this regard [6, 93, 94]. Since, these studies do not require fake training videos, they have the ability to generalize to invisible attacks. According to the reviews Table 5 on face manipulation, there are several trends that researchers can pay attention to. Many researchers [50, 93, 95] used unsupervised GAN-based frameworks for translations and manipulations of digital

Table 5 Review of different types of Deepfake generation and GANs techniques along with their Datasets and Github code

Study	Gan Type	Method	Dataset	Accuracy	Code Link
[50]	STGAN, PGGAN & StyleGAN	DNN (VGG-Face3 with ResNet50)	Own dataset (CelebA, FFHQ)	84.7%, 90.2%	–
[51]	PGGAN	CNN	Own dataset (CelebA)	80.88%	https://github.com/tadax/pggan
[52]	FSGAN	(R.N.N.) & (CNN)	FaceForensics++ & Face2 Face	99.4%	https://github.com/YuvalNirkin/fsgan
[41]	MaskGAN	Dense Mapping Network (D.M.N.)	Own date set (CelebAMask-HQ)	77.3%	https://github.com/switchablenorms/CelebAMask-HQ
[39]	StarGAN v2	CNN	Own dateset (CelebA-HQ)	92.1%	https://github.com/clovaai/stargan-v2
[54]	L2M-GAN	CNN	Own dateset (CelebA-HQ)	95%	https://github.com/songquanpeng/L2M-GAN
[1]	FSGAN	CNN	F++ (Face2 Face)	100%	https://github.com/ondyari/FaceForensic

images, and they got a high accuracy of close to 100%. The practical side of facial manipulation, the resource code of most GAN approaches, is publicly available. So, we need to work in other fields without using the GAN architecture. There are few resources for fake datasets that are publicly accessible for study in this field. Most studies create a dataset with their own GANs to test the method's effectiveness. This causes an unfair comparison due to many different factors. Most of the experiments that were conducted under unseen conditions, such as noise, blur, cropping, and compression, hindered the work of the detector, thus increasing the error rate. Nevertheless, there are datasets with abundant taxonomic information, especially for face attribute manipulation. For example, CelebA and CelebAHQ are the most widely used real-face datasets for generating attribute manipulation, as shown in Fig. 8. Another recent study by Alkishri et al. [96] explores the challenges of detecting GAN-generated images, particularly when using adversarial techniques to remove or modify GAN fingerprints. The authors analyze high-frequency Fourier modes, highlighting that GAN-generated images have systematic shortcomings in replicating these features. Their study shows that by eliminating GAN traces, the detection rate of fake images can be reduced by up to 50%, emphasizing the weakness of current deepfake detection methods and the need for more robust algorithms.

7.2 Challenges facing deepfake detection

In Deepfakes detection methods, there are many challenges that restrict precision levels.

7.2.1 Degradation of quality

In order to share data over the Internet, there will be a reduction in quality due to compression and resizing, so that it can suffer from noise and so on. In turn, Deep-fake detection algorithms may have difficulty detecting this fake data. It was shown in the FaceForensics++ paper that forged faces were more easily identified by Face Forensics++ than by other methods. Despite being able to classify raw input data correctly, the methods were not effective on compressed video. To reduce counterfeiting, it is important to develop deep-fake detection methods that are robust to degraded quality and content.

7.2.2 Artifact and ganprintr

The omission of fingerprints or other details, such as artifacts, may be used to fool Deepfake detectors. The GANprintR paper, for example, outlines a simple method for extracting GAN fingerprints from synthetic fake images based on auto-encoder conversion. This technique uses facial distortion detection systems to preserve visual consistency. Some deep-fake detection methods can be compromised by methods such as these. Furthermore, suppose Deepfake detection techniques specialize in particular characteristics, such as an irregular blinking rate in a video. In that case, individuals might try to invent improved Deepfake methods of production that do not have those characteristics [97–99].

7.2.3 Ineffective generalization and development of deep fake technology

Several Deepfake detection methods cannot generalize well to a variety of content types and manipulation techniques. A total of 46 studies revealed that in one of their experiments that they used their 3D CNN detectors with cross-manipulation techniques on actual videos and fake videos that were produced using three methods. As compared to other experiments, the trained model was less effective at detecting deepfakes generated with a technique on which it wasn't trained. This ability to generalize across unidentified domains is important in Deepfake detection. Implementing deepfake detection in the real world would likely not reveal the true source and type of deception applied to a particular situation, such as detecting misinformation. It will become increasingly important to update strategies that are poorly generalized to cover these new deep-faked approaches as new deep-faked development methods become available. Some Deepfake detection techniques may indeed be able to generalize to unknown domains more easily than others. It is necessary to define specific strategies and establish detection methods that could adapt to unconstrained conditions.

8 Conclusions

This review gives a comprehensive study and extensive analysis of the research papers on the topics of Deepfake generation using GANs approaches and Deepfake detection by utilizing more than 100 research publications. The taxonomy of Deepfake generating methods based on GANs, popular datasets details, and the categorization of various Deepfake detection approaches with their challenges are described in this paper. A synthetic media form is known as Deepfakes (a combination of the terms “deep learning” and “fake”). Deepfakes involve the replacement of a person’s likeness in an existing photograph or video with another individual. Deepfakes utilize strong machine learning and artificial intelligence techniques to edit or generate visual and audio content with a high potential to deceive users. The majority of face manipulations used today can be identified in controlled contexts, particularly if fake detectors are tested under the same circumstances in which they were developed. Most of the benchmarks used in this review study attained extremely low mistake rates in manipulation detection. This scenario does not come close to being realistic because fake photos and movies are frequently shared on social networks with a great deal of variance, such as compression, resizing, and noise. Additionally, most researchers used their dataset, which caused unfair comparisons with other studies.

Techniques for manipulating facial expressions are also constantly improving. Considering these factors, further research into the generalization abilities of the fake detectors should be conducted. GANs (generative adversarial networks) modeling is an unsupervised machine learning technique that involves automatically detecting and learning patterns in input data so that the model may be used to produce new examples that could have been derived from the original data. The popular GANs were reviewed based on the type of Deepfake face manipulation. However, GAN training continues to be a problem. Generated images lack realism due to general face information, such as symmetry. The created photos lack authenticity because of a lack of facial information, such as symmetry. In whole-face synthesis, fraudulent images are very convincing, making it hard to identify real from fake. Current research focuses on training stability, resolution, and adjustable face features. The use of GAN as an identification technique for fake images achieved superior results. However, simultaneously, it discovers the fake image through a systematic and procedural process. Therefore, what happens if we remove the GAN from the fake images? The current approaches will be able to discover fake images easily by using a GAN. So, we need to propose new computation techniques to discover fake images.

We believe that this review has been organized to provide a structured analysis of important research papers and to discuss each study’s main findings and conclusions. Our investigation reveals shortcomings in manipulation detection benchmarks due to real-world scenario variations and biased dataset comparisons. Current research priorities revolve around enhancing GAN training stability, resolution, and manipulable facial features. Moreover, GANs have shown superior results in identifying fake images; however, their reliance prompts a systematic approach to detecting fakes. This dependency raises questions about detecting fake images with or without manipulated GAN architecture, urging the need for novel computational techniques to identify manipulations without GAN assistance.

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Data availability The datasets generated and/or analyzed during the current study are publicly available in the Github repository and can be accessed at [<https://github.com/>]. Moreover, the datasets used within the experimental work are publicly available online.

Declarations

Ethics approval and consent to participate Not applicable. The study conducted in this paper was completed on real datasets that are publicly available on the GitHub repository. We have some of these datasets in Table 5.

Consent for publication Not applicable.

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